

Aim:- To write SQL statements for implementing ALTER, UPDATE & DELETE operations on database table

Theory:- In DBMS, SQL provides commands to modify database structure & data.

1. ALTER Command-

- Used to modify the structure of a table
- Can add, delete or modify columns.

Ex Add column
Modify datatype
Drop column

2. Update Command

- Used to modify existing data in a table
- Works with WHERE clause to update specific records

3. DELETE Command

- Used to remove records from a table.
- Can delete specific rows or all column rows.

These commands are part of Data Definition Language (DDL) & Data Manipulation Language (DML) & are essential for maintaining & managing the Database.

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Algorithm

1. Start SQL environment
2. Select the database using USE
3. Apply ALTER command on existing tables.
4. Update records using UPDATE command.
5. Delete records using DELETE command
6. Display results using SELECT.C

Program

USE Collage DB

ALTER TABLE Student
ADD Address VARCHAR(200);

ALTER TABLE Student
MODIFY Phone_No VARCHAR(15);

ALTER TABLE Student
DROP COLUMN Address;

UPDATE Student
SET Name = 'Amit'
WHERE Student-ID = 1;

UPDATE Student
SET Semester = 5
WHERE Student-ID = 2;

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Output

Before Update/Delete

Student_ID	Name	Semester
1	Rahul	3
2	Priya	4
3	Aman	2

After update/Delete

Student_ID	Name	Semester
1	Amit	3
2	Priya	5

```
UPDATE Course
SET Tuition_fee = 50000
WHERE Course_ID = 101;
```

```
DELETE FROM Student
WHERE Student_ID = 3;
```

```
DELETE FROM Exam;
```

```
SELECT * FROM Student;
SELECT * FROM Course;
```

Result Thus the SQL commands ALTER, UPDATE, DELETE were successfully implemented on the database tables.

Viva Questions:-

Q1 What is the use of ALTER command?

Ans ALTER command is used to modify the structure of any existing table.

Q2 What is the purpose of UPDATE Command?

Ans UPDATE command is used to modify existing records in a table.

Q3 What is DELETE Command?

Ans DELETE command is used to remove records from a table.

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Q4 What happens if WHERE clause is not used in UPDATE or DELETE?

Ans All records in the table will be updated or deleted.

Q5 Difference between DELETE & DROP?

Ans DELETE removes records from a table.

DROP removes the entire table structure & table-data

Aim:- To write SQL queries to implement different types of JOINS in database.

Theory:- A Join in SQL is used to combine data from two or more tables based on a related column between them (usually primary key & Foreign key).

Types of JOINS:- A JOIN in SQL is used to combine data from two or more tables based on a related column between them.

1. INNER JOIN - Returns only matching records from both tables.

2. LEFT JOIN (LEFT OUTER JOIN):-

Returns all records from left table & matching from right table.

Non - matching values $\rightarrow$ NULL

3. RIGHT OUTER JOIN:-

Returns all records from right table & matching from left - table.

4. FULL JOIN:- Returns all records from both tables (matching + non-matching)

Examples

Student table	these are connected
Course table	via Course-id
Subject table	Teacher-id
Teacher table	student-id.
	Subject-id.

Algorithm:-

1. Start SQL environment
2. select the database
3. Identify related tables
4. Apply different types of JOIN queries
5. Display results

Program -

Step 1 Select Database
use CollegeDB;

INNER JOIN

Step 2 Student & Course (Match)

SELECT Student.student-ID, Student.Name,
Course.Name AS Course-Name

FROM Student

INNER JOIN Course

ON Student.course-id =

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Course.course-id

LEFT JOIN

Step 3: Student & Course (All Student)

```
SELECT student.Student_ID, student.Name,  
Course.Name AS Course_Name  
FROM Student  
LEFT JOIN Course  
ON Student.Course_ID = Course.Course_ID;
```

RIGHT JOIN

Step 4: Course & Student (All Course)

```
SELECT Course.Course_ID, Course.Name,  
Student.Name  
FROM Student  
RIGHT JOIN Course  
ON Student.Course_ID = Course.Course_ID;
```

JOIN with teacher & Subject

```
SELECT Subject.Name AS Subject_Name,  
Teacher.Name AS Teacher_Name  
FROM Subject  
INNER JOIN Teacher  
ON Subject.Subject_ID = Teacher.Teacher_ID;
```

<u>output</u>		
<u>Student_Name</u>	<u>Course_Name</u>	<u>Subject_Name</u>
Rahul	B.tech	DBMS
Priya	B.tech	DS

Multiple Join

Step 6 : Student + Course + Subject

```
SELECT Student_Name, Course_Name As
Course_Name, Subject_Name As Subject_Name
FROM Student
```

```
INNER JOIN Course ON Student.Course-ID =
Course.Course-ID;
```

```
INNER JOIN Subject ON Course.Course-ID =
Subject.Course-ID;
```

Display output →

SELECT * FROM Student;

SELECT * FROM Course

Results Thus, Different types of JOIN operation
(INNER, LEFT, RIGHT) were successfully
implemented using SQL query.

Viva Questions -

Q1. What is a JOIN?

Ans A Join is used to combine data from
two or more tables based on a related
column

Q2 Difference between INNER JOIN & LEFT JOIN?

Ans • INNER JOIN → only matching records
- LEFT JOIN → all records from left + matching from right.

Q3 What is RIGHT JOIN?

Ans Returns all records from the right tables & matching from the left.

Q4 What is the use of ON Clause?

Ans It specifies the condition for joining table.

Q5 What happens if no matching record exists?

Ans NULL values are returned (in LEFT/RIGHT JOIN)